the antibodies are secreted by the $CD11c^+$ or $CD11c^-$ cells in the $CD27^-CD43^+$ population and furthermore, whether this population is a real ASC population or an activated $CD27^-$ memory B-cell population that is on its way to plasmablast differentiation and therefore might be considered to be an ASC precursor.

In conclusion, there are still a lot of unknowns about the human B-cell response and the various stages through which B cells go to become either a memory B cell or an ASC. Specifically, the bisection point(s) after which differentiating B cells are destined to become memory B cells or ASCs remains a topic for further investigation. Recent advances in multiparameter flow cytometry analysis and single-cell RNA sequencing, together with the possibility of analyzing the development of antigen-specific B cells (eg, after severe acute respiratory syndrome coronavirus 2 [SARS-CoV-2] infection or vaccination), will likely advance our understanding of human B-cell differentiation dynamics beyond the major (more conventional) populations.[4] It is hoped that these studies will also provide insight into how $CD11c^+$ populations relate to one another and to $CD11c^-$ populations and also whether expression of certain markers are reflecting merely a specific activation state rather than a separate B-cell population.

DISCLOSURE STATEMENT

Disclosure of potential conflict of interest: The authors declare that they have no relevant conflicts of interest.

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Reply to Dages et al: You AIn't using it right—artificial intelligence progress in allergy

To the Editor:

We (ChatGPT and I) read with interest the letter by Dages et al regarding the limitations of the artificial intelligence (AI) tool ChatGPT.[1] We concur that the tool did not accurately address their query on cefazolin administration in a patient with penicillin allergy. However, we believe that this outcome reflects the current limitations of the tool's use as a search engine rather than limitations of the AI model itself. We fully agree that the submission by Dages et al[1] is typical and acknowledge that results can often deviate convincingly from what is expected. Thus, the report by Dages et al[1] is an important cautionary example.

Because language models generate responses based on data likelihood and not factual accuracy, it is important that we understand their function so that we can use them effectively.

To demonstrate, we used ChatGPT v4 for a more complex approach to answer the same question asked by Dages et al.[1] We propose that one must (for now) explicitly pose queries such as, "Can we automate downloads from PubMed, analyze the content, and interpret the result concerning the safety of cefazolin for patients with penicillin allergy?"

We began by asking, "Which R package is able to query PubMed and return abstracts?" (R being our programming language of choice). ChatGPT automatically generated R code that could gather relevant abstracts from PubMed by using our key words. From here, we incrementally asked for additional code to process and interpret the information.

The full code set provided us with an R computer program, written by ChatGPT, that performs the automated PubMed queries relative to the question and saves all abstracts and references. It then performs term frequency and correlation analysis on the texts. Lastly, the text corpus is used as input for ChatGPT's interpretation of the original question to return a response based on the referenced summaries. The result intuitively demonstrates, to the human reader, what the text-based information consisted of before being interpreted by ChatGPT's model.

The final result consists of Fig 1, illustrating the summarized content from PubMed (Fig 1, *A-D*), a balanced text report based on those summaries (Fig 1, *E*), and the list of publications used in the analysis (Fig 1, *F*).

The supporting data were automatically sourced for the first 20 PubMed results, as well as the Uniform Resource Locators (URLs) and PubMed identifiers (PMIDs) printed and automatically imported into a citation manager. We limited the method to the default of 20 publication abstracts so that it would be easy to replicate; however, using the ChatGPT application programming interface (API) allows the system to be automated, with relative ease, to run on far larger data sets.

Our findings show that AI can provide valuable information for problem solving. We believe that the concerns expressed by Dages et al[1] are justified because most users initially assume that responses to queries are supposed to be fact based. By adjusting the approach, we can use AI tools to improve how we find and analyze information.

However, we must highlight the fact that as it stands, the process is time-intensive—our session required 2 hours. It would be unrealistic to expect every user to devote such effort. We anticipate the advent of more user-friendly tools and interfaces, making data retrieval and analysis accessible and efficient.

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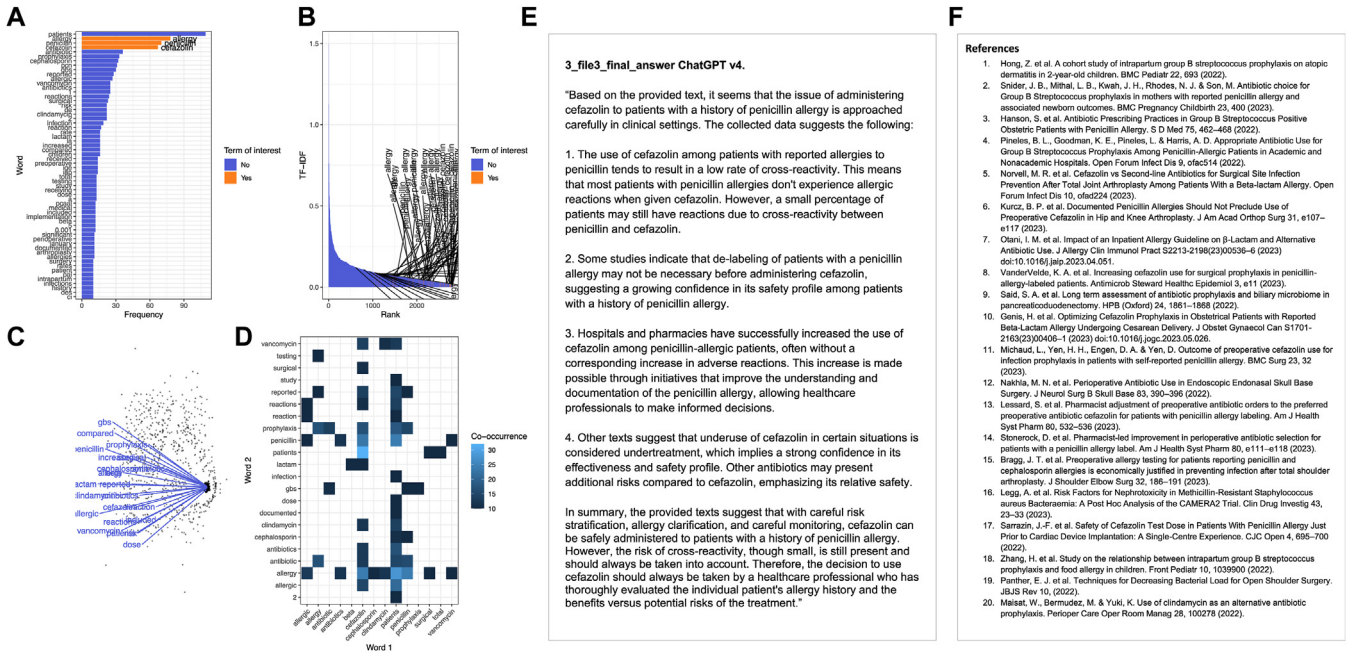


FIG 1. Term frequency in automated PubMed queries and final interpretation. **A,** The most common words in abstracts related to cefazolin and penicillin allergy (quantile 0.95 is shown). **B,** Term frequency–inverse document frequency (TF-IDF) for terms of interest in abstracts related to the search terms "cefazolin" and "penicillin" "allergy" (text labels highlight matches to the 3 terms). **C,** Network plot of edge weights based on the frequency of co-occurrence in abstracts related to cefazolin and penicillin allergy (quantile 0.98 is shown). **D,** Heatmap of term co-occurrence in abstracts (co-occurrence threshold of >9 is shown). This same data set was used throughout for subsequent interpretation by ChatGPT. **E,** Final response by ChatGPT based on summaries from abstracts. **F,** References used for ChatGPT's final response.

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Reply

To the Editor:

We appreciate and agree with Dr Lawless's main points—that the results of artificial intelligence (AI) programs such as Chat GPT are highly dependent on specific prompts and newer versions of AI are less likely to create errors.[1] The aim of our letter,[2] however, was to provide a simple example of AI while highlighting some of the possible risks or inaccuracies of Chat GPT. Models do not know what they do not know, and errors such as hallucinations are a known risk. Therefore, as Dr Lawless points out, a carefully designed query, in addition to expertise with AI, is absolutely necessary to minimize such errors. We also agree with Dr Lawless that AI is a powerful tool that can revolutionize medicine and improve health care. However, as with any tool, caution is advised—the risk of error is never zero.

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